

Zain Siddiqui

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1 PERSONAL BACKGROUND

Born December 16, 1995 in Cleveland, TN, USA.

EDUCATION

Washington University in St. Louis, St. Louis, MO Enrolled: Fall 2021— Expected Graduation: Spring 2026

PhD in Mathematics

Thesis Advisor: Renato Feres

Georgia Institute of Technology

Fall 2014-Spring 2019

BS in Mathematics

RESEARCH INTERESTS

Some of my interests are dynamics, stochastic processes, and ergodic theory.

In particular, I am interested in dynamical billiards. Specifically, I am focused on polygonal billiards and the topology of their unfoldings, and I wish to understand these unfoldings using both topological and stochastic methods.

PUBLICATIONS

Don't rush me!

TEACHING EXPERIENCE

Washington University in St. Louis

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| • Linear Algebra Lecturer | Summer 2025 |
| • Directed Reading Program, <i>Markov Chains and Mixing Times</i> | Fall 2024 |
| • Calculus I Teaching Assistant | Fall 2024 |
| • Calculus III Lecturer | Summer 2024 |
| • Directed Reading Program, <i>Geometry: Euclid and Beyond</i> | Spring 2024 |
| • Calculus I Teaching Assistant | Spring 2024 |
| • Calculus II Teaching Assistant | Fall 2023 |
| • Calculus II Teaching Assistant | Spring 2023 |

Georgia Institute of Technology

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| • Calculus III Teaching Assistant | Spring 2019 |
| • Calculus III Teaching Assistant | Fall 2018 |
| • Calculus III Teaching Assistant | Spring 2018 |
| • Calculus III Teaching Assistant | Fall 2017 |
| • Calculus III Teaching Assistant | Fall 2016 |

TALKS

- Szegő Seminar, Washington University in St. Louis, *What an Incidence!/: Hilbert's Axioms in Geometry* - March 2024
- Candidacy Talk, Washington University in St. Louis, *Unfolding a Billiard Table* - March 2024

- Lightning Talk, St. Louis Topology Conference, *Unfolding a Billiard Table* - May 2024
- Szegő Seminar, Washington University in St. Louis, *What's a Foliation?* - September 2024
- Szegő Seminar, Washington University in St. Louis, *A Meditation on the Universal Cover* - October 2025
- Geometry and Topology Seminar, Washington University in St. Louis, *Singular Manifolds Foliated by Graphs* - October 2025
- Szegő Seminar, Washington University in St. Louis, *The Generator of Brownian Motion* - February 2026

SERVICE

- Organizer at St. Louis Topology Conference 2024